

Short Communication

Effect of vanadium addition on the microstructure and properties of AlCoCrFeNi high entropy alloy

Yong Dong, Kaiyao Zhou, Yiping Lu^{*}, Xiaoxia Gao, Tongmin Wang^{*}, Tingju Li

School of Materials Science and Engineering, Dalian University of Technology, Dalian 116024, China

ARTICLE INFO

Article history:

Received 14 November 2013

Accepted 18 December 2013

Available online 27 December 2013

ABSTRACT

The purpose of this study is to investigate the effects of vanadium addition on the microstructure and mechanical properties of AlCoCrFeNiV_x (x values in molar ratio, $x = 0, 0.2, 0.5, 0.8, 1.0$) alloys. All the alloys were found to display a crystalline structure of simple body centered cubic (BCC). For AlCoCrFeNi and AlCoCrFeNiV_{0.2} alloys, Cr and Fe elements segregated to the center of grain while Al and Ni elements segregated to the rest areas. With the increase of V content exceeding to $x = 0.5$, the homogenized polycrystalline grain can be obtained. For AlCoCrFeNiV_{0.2} alloy, the compressive strength and plastic strain were as high as 3297.8 MPa and 26.8%, respectively, which were rare in high entropy alloys to date. The fine nanoscale spinodal decomposition microstructure was a key factor for the high fracture strength of AlCoCrFeNiV_{0.2} alloy. The values of Vickers hardness increased from HV534 to HV648.8 with the increase of V content. The solid-solution strengthening of the body centered cubic matrix was found as the main factor that strengthened the alloys. With the increase of V contents from $x = 0$ to $x = 1.0$, the transformation of ferromagnetic behavior to paramagnetic behavior takes place.

© 2013 Elsevier Ltd. All rights reserved.

1. Introduction

Recently, a new type of alloys, i.e. high entropy alloys (HEAs) also known as multi-component equimolar or near-equimolar alloys, are becoming new research hotspots in the metallic materials community [1,2]. Compared to the conventional metallic alloys based on one or two major elements, HEAs generally contain 5–13 principal elements with the concentrations of each of them being between 5 and 35 at.% [2,3]. The previous studies [4–6] have shown that HEAs predominantly consist of a simple face centered cubic (FCC), body centered cubic (BCC) or FCC+BCC structure solid solution phase owing to the high entropy of mixing, instead of many intermetallic phases or other complex phases. However, a small quantity of intermetallic compound phase/metastable particles also can form in some HEAs. Then Zhang et al. and Liu et al. [7–10] studied the effect of the atomic-size difference (ΔR), the mixing entropy (ΔS), the mixing enthalpy (ΔH) and valence electron concentration (VEC) on the formation and stability of solid solution phases of HEAs, which successfully predicted the formation and stability of solid solution phases. Moreover, numerous studies also focused on the promising properties of HEAs in structure materials, such as high hardness [11,12], high strength [13,14], high wear resistance [15,16], excellent corrosion resistance [17,18], and good temperature stability [19,20]. Take the

AlCoCrFeNiTi_{0.5} [13] and Al_{0.2}Co_{1.5}CrFeNi_{1.5}Ti alloys [15] for example, the yield strength, fracture strength and plastic strain of AlCoCrFeNiTi_{0.5} alloy are as high as 2.26 GPa, 3.14 GPa, and 23.3%, respectively, which are superior to most of the high-strength alloys such as bulk metallic glasses. The Al_{0.2}Co_{1.5}CrFeNi_{1.5}Ti alloy [15] has the similar hardness with SUJ2, but the wear resistance of the former is 3.6 times that of the latter.

In the past, the researchers focused on Al_{0.5}CoCrCuFeNi alloy [21,22], AlCoCrCuFeNiMo_x [23], Al_{0.5}CoCrCuFeNiTi_x [24], Al_{0.5}CoCrCuFeNiB_x [25], and Al_{0.5}CoCrCuFeNiV_x [26] alloys. Later, the Cu element that is segregated in interdendrite guides the focus from Al–Co–Cr–Cu–Fe–Ni to Al–Co–Cr–Fe–Ni alloy systems. The microstructure and properties of AlCoCrFeNiMo_x [12], AlCoCrFeNiTi_x [13], AlCoCrFeNiC_x [27], and AlCoCrFeNiSi_x [28] alloys have been reported intensively. Thereinto, high entropy AlCoCrFeNi alloy [13] with BCC structure has high compression strength (2.83 GPa) and plastic strain (26.9). Element vanadium possesses the same structure with the AlCoCrFeNi alloy and is often added to alloys as the element of solution strength to add to alloys, such as in Al_{0.5}CoCrFeNiTiV_x [26], K417G superalloy [29], and TiZrNb–MoV_x alloys [30]. In the Al_{0.5}CoCrCuFeNiV_x alloy system [26], with the increase of V contents from $x = 0$ to $x = 3.0$, the crystal structure of the alloys transforms from FCC to FCC+ σ phase, then to BCC structure, which shows that the V element can stabilize the BCC structure. In this paper, a novel alloy system AlCoCrFeNiV_x ($x = 0, 0.2, 0.5, 0.8, 1.0$) was designed and the variation of its microstructure and properties were investigated.

^{*} Corresponding authors. Tel.: +86 411 84709400; fax: +86 411 84708940.

E-mail addresses: luyiping@dlut.edu.cn (Y. Lu), tmwang@dlut.edu.cn (T. Wang).

2. Experimental details

The AlCoCrFeNiV_x (when $x = 0, 0.2, 0.5, 0.8$, and 1.0 , the alloys were denoted as V0, V0.2, V0.5, V0.8, and V1.0, respectively) alloys were prepared by vacuum arc melting. The alloys were repeated melting at least five times with electromagnetic stirring for homogeneity. Then the alloys were directly solidified in a water-cooled cold copper hearth and the solidified ingots were about 30 mm in diameter and 15 mm in thickness. The details of the melting process can be found in Ref. [31]. The samples (15 mm in diameter and 10 mm thick) which were taken from the as-cast ingots were polished and etched with alcohol dilute aqua regia for examining the microstructure under a scanning electron microscope (SEM, Zeiss supra 55) with energy dispersive spectrometry (EDS). And the crystalline structures of as-cast samples were characterized by the X-ray diffraction (XRD, EMPYREAN with Cu radiation target) with the 2θ scan from 20 to 100 deg. For the compressive test (test standards used ASTM: E9-09), specimens with diameter of 5 mm and length of 10 mm were prepared. And the test was conducted at a rate of 3 mm/min. Hardness measurements of the alloys (test standards used ASTM: E384-11e1) were conducted using a Vickers hardness tester (MH-50) under a load of 1000 g for 15 s. Five measurements were made for each sample to obtain the averaged experimental data. The magnetization curves were measured with JDM-13T vibrating sample magnetometer.

3. Results and discussion

3.1. X-ray diffraction analysis

The X-ray diffraction patterns of the AlCoCrFeNiV_x alloys are shown in Fig. 1a. All alloys exhibit the reflections of a main single body centered cubic solid solution structure, which indicates the addition of V element to AlCoCrFeNi alloy has no effect on the matrix crystalline structure. The ordered peak around 30.7° appears in V0.5 alloy, which indicates the appearance of a small quantity of ordered phase. Similar experimental result was reported in Ref. [22]. Fig. 1b shows the detailed scans for the highest peak of (110) of BCC solid solution phase. The peak of (110) shifts to the lower 2θ with the increase of V. This change is identical to the variation of the lattice constants of the BCC phase, which are calculated with the strongest peak of (100). The values of the lattice constant are 0.28755, 0.28812, 0.28895, 0.28915, and 0.28916 nm in the AlCoCrFeNiV_x alloys with $x = 0, 0.2, 0.5, 0.8$, and 1.0 , respectively. It is found that the lattice constants increased with the increase of V element. This indicates that the solution limit of the V element in AlCoCrFeNi alloy is more than 16.67 at.%, which corresponds to $x = 1.0$. The atomic radius of the experiment elements is summarized in Table 1. The V element has the second largest atomic radius in the six elements, which would lead to the lattice distortion that corresponds to the increased lattice constant.

According to the solid solution rules for HEAs [7–10], besides high entropy of mixing (ΔS), other factors can also influence solid solution phase formation such as the atomic size differences (ΔR), the enthalpy of mixing (ΔH), the electronegativity differences (ΔX), and the valence electron concentration (VEC). The parameters of ΔS , ΔH , ΔR , VEC and ΔX for AlCoCrFeNiV_x alloys are listed in Table 2. The calculation formulas for ΔS , ΔH , ΔR , VEC and ΔX can be obtain from Refs. [7,8]. The needed physicochemical/ther-

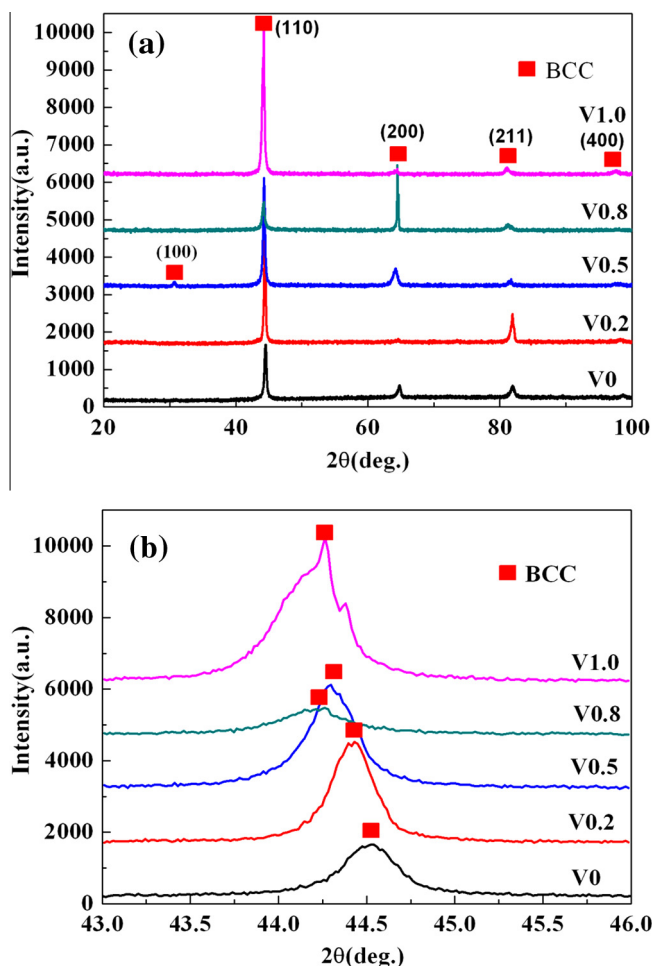


Fig. 1. (a) XRD patterns of AlCoCrFeNiV_x alloys, (b) the detailed scans for the peak of (110) of BCC solid solution phase.

Table 1

Properties of the experimental elements (from the periodic table software).

Element	Atomic radius (pm)	Pauling electronegativity	VEC
Al	143	1.61	3
Co	125	1.88	9
Cr	128	1.66	6
Fe	126	1.83	8
Ni	124	1.91	10
V	134	1.63	5

Table 2

Parameters of ΔS , ΔH , ΔR , VEC, and ΔX for AlCoCrFeNiV_x alloys.

Alloy	ΔR	VEC	ΔX	ΔH (kJ mol ⁻¹)	ΔS (J K ⁻¹ mol ⁻¹)
V0	5.438	7.2	0.1206	-12.32	13.38
V0.2	5.372	7.12	0.1216	-13.08	14.22
V0.5	5.276	7	0.1226	-13.95	14.7
V0.8	5.182	6.9	0.123	-14.58	14.87
V1.0	5.122	6.83	0.1231	-14.89	14.9

Table 3

Chemical mixing enthalpies of the atomic pairs among the alloying elements (unit: kJ/mol) [32].

Element	Al	Co	Cr	Fe	Ni	V
Al	-	-	-	-	-	-
Co	-19	-	-	-	-	-
Cr	-10	-4	-	-	-	-
Fe	-11	-1	-1	-	-	-
Ni	-22	-7	-7	-2	-	-
V	-16	-14	-2	-7	-18	-

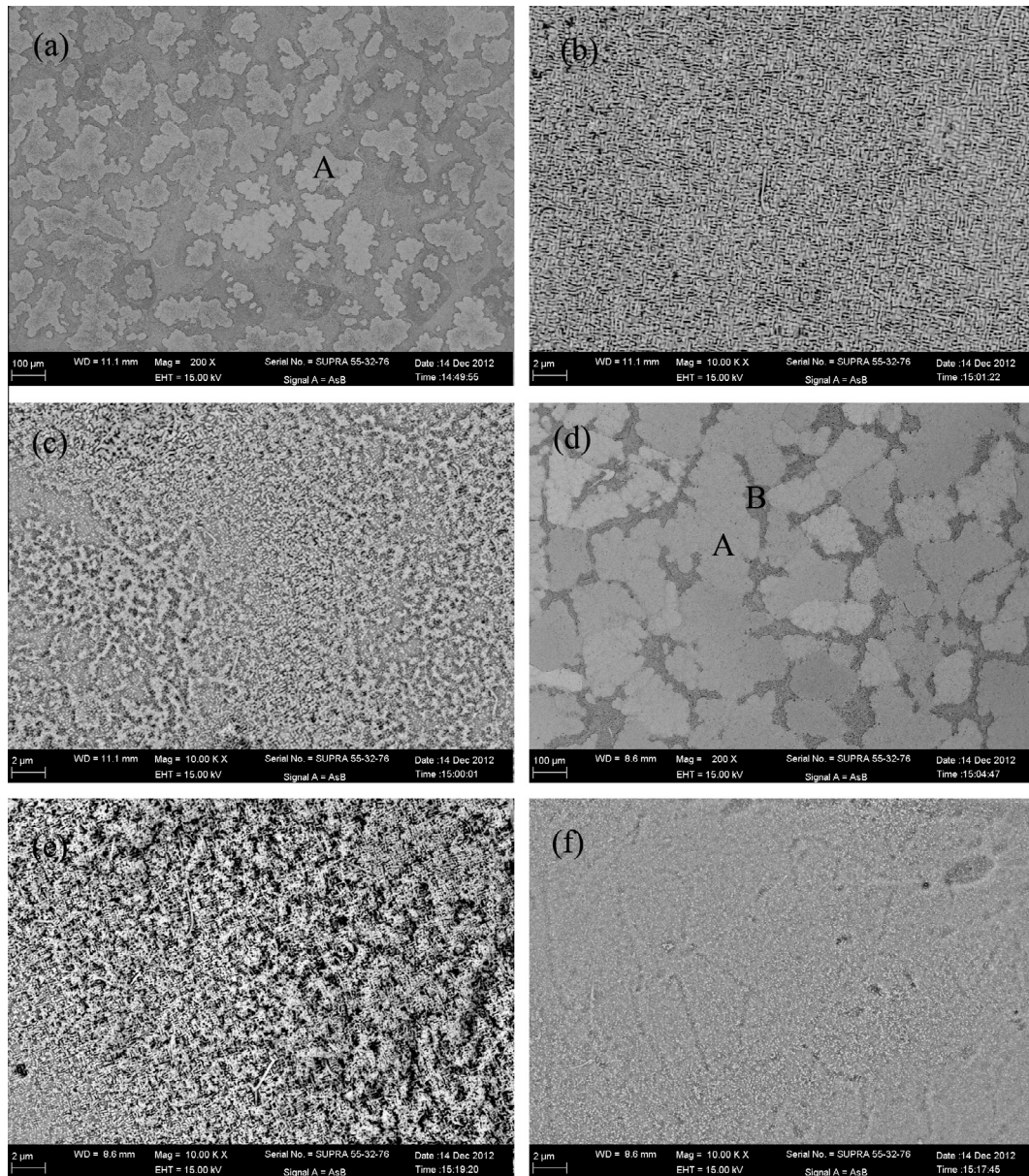


Fig. 2. SEM backscattered electron images of V0 and V0.2 alloys, (a) V0 alloy, (b) and (c) are the magnification of areas B and A in (a), respectively. (d) V0.2 alloy, (e) and (f) are the magnification of areas B and A in (d), respectively.

Table 4

Chemical composition of different regions in atomic percentage.

Alloy	Regions	Al	Co	Cr	Fe	Ni	V
V0.0	A	23.8	19.8	16.5	17.6	22.2	–
	B	1.1	19.0	47.2	28	4.7	–
	Nominal	20	20	20	20	20	–
V0.2	A	20.5	19.5	18.4	18.3	19.6	3.7
	B	0.9	17.3	44.5	27.3	4.8	5.2
	Nominal	19.23	19.23	19.23	19.23	19.23	3.85
V0.5	DA	17.6	18.3	18.9	18.2	17.9	9.1
	LA	16.8	18.2	19.6	18.4	17.7	9.3
	Nominal	18.18	18.18	18.18	18.18	18.18	9.1
V0.8	DA	17.7	16.4	18.9	17	15.8	14.2
	LA	17.3	16.8	18.3	17.2	16.4	14
	Nominal	17.24	17.24	17.24	17.24	17.24	13.8
V1.0	DA	16	16.4	17.6	16.7	16.2	17.1
	LA	16.3	16.4	17.5	16.7	16.2	17
	Nominal	16.67	16.67	16.67	16.67	16.67	16.66

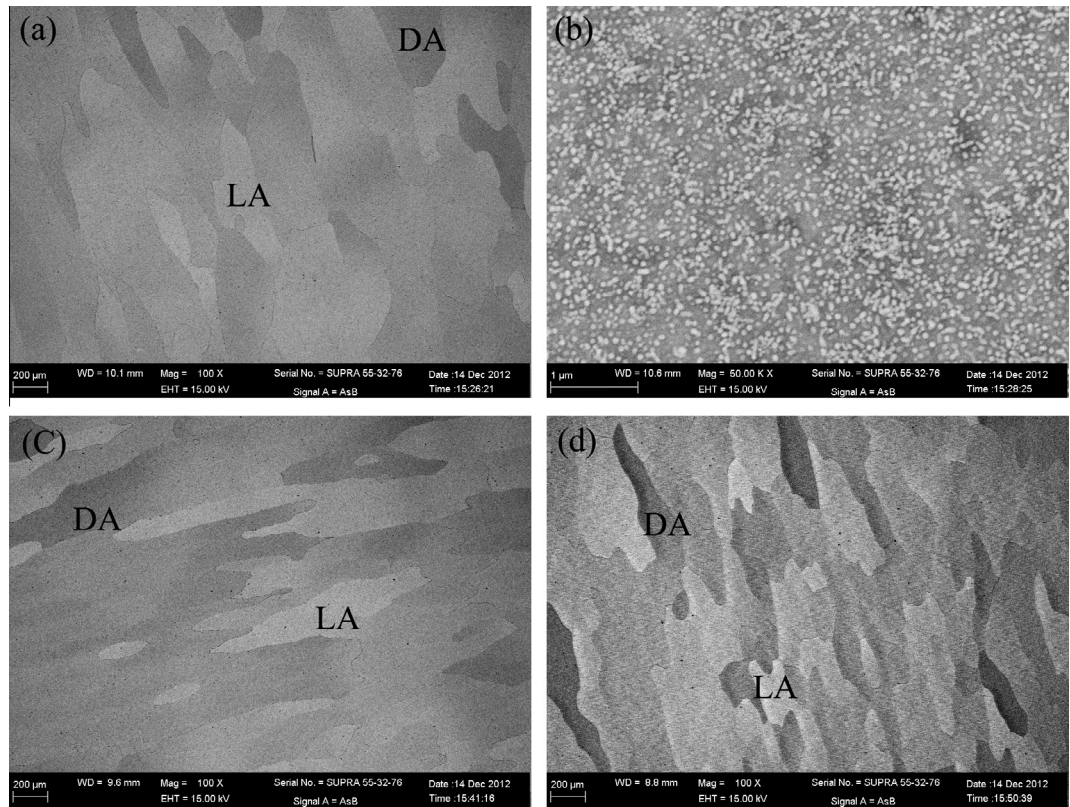


Fig. 3. SEM backscattered electron images of V0.5, V0.8, and V1.0 alloys, (a) V0.5 alloy, (b) high-magnified image of (a), (c) V0.8 alloy, (d) V1.0 alloy.

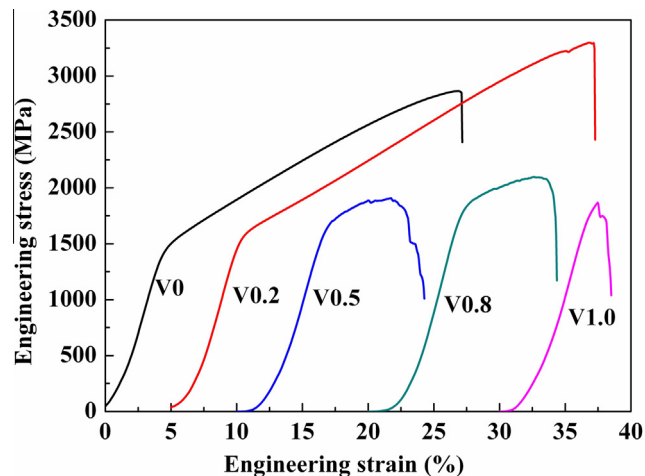


Fig. 4. Compressive engineering stress–strain curves of AlCoCrFeNiV_x alloys.

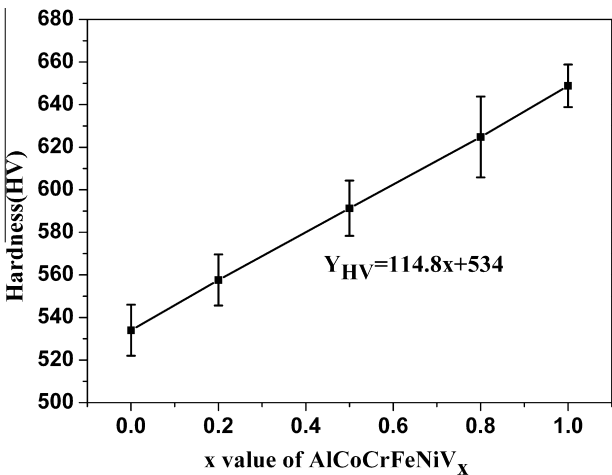


Fig. 5. Vickers hardness curve of AlCoCrFeNiV_x alloys.

Table 5
Mechanical properties and Vickers hardness of AlCoCrFeNiV_x alloys.

Alloy	σ_y (MPa)	σ_b (MPa)	σ_p (%)	HV
V0	1378.9	2864.9	22.7	534
V0.2	1492.4	3297.8	26.8	557.6
V0.5	1597.7	1907.2	5.1	591.3
V0.8	1695.8	2097.4	5.7	624.8
V1.0	1726.5	1867.1	0.7	648.8

modynamic parameters of experimental elements for the calculation are also listed in Tables 1 and 3 [32]. As can be seen from the results, with the increase of V contents, the atomic size

differences, the enthalpy of mixing, and the valence electron concentration of the AlCoCrFeNiV_x alloys decreased while the electronegativity differences and the entropy of mixing increased. Apparently, any single parameter could not successfully predict the formation of solid solution. Although the decrease of ΔH (from -12.32 to -14.98 kJ mol⁻¹) and the increase of ΔX (from 0.1206 to 0.1231) may lead to the appearance of intermetallic compounds, the decrease of the VEC and ΔR in this alloy system (from 7.2 to 6.83) could stabilize BCC phase structure. Therefore, the factors, such as ΔS , ΔH , ΔR , ΔX and VEC, need to be taken into account when predicating on the stability of solid solution phases in high entropy alloys.

3.2. Microstructure

Fig. 2 presents SEM backscattered electron images of V0 and V0.2 alloys. Intragranular dendritic segregation clearly appears in the two alloys. The dendritic segregation area and interdendritic segregation area are denoted as A and B, respectively. The detailed SEM images of V0 alloy for area A and B are shown in Fig. 2c and b, respectively. The area A and B showing in Fig. 2b and c and Fig. 2e and f has been shown to be nanoscale spinodal microstructure [33,34]. According to the previous research [33], the bright section and the dark section of interconnected microstructure were composed of (Fe, Cr)-rich element and (Ni,Al)-rich element whereas section A has a higher amount of Al and Ni than section B. Table 4 shows the chemical composition of different areas analyzed by EDS. In V0 alloy, area A was enriched with Al and Ni elements while area B was enriched with Fe and Cr elements. V0.2 alloy has a similar microstructure with that of V0 alloy, as shown in Fig. 2d. Compared to V0 alloy, the proportion of area A increased, evidenced in Fig. 2e. And the same as V0 alloy, Cr and Fe elements segregated to area B while Al and Ni elements segregated to area A. Fig. 3 presents SEM backscattered electron images of V0.5, V0.8, and V1.0 alloys. When the value of x is equal to or greater than 0.5, no macroscopic intragranular dendritic segregation took place and a microstructure of nanoscale spherical precipitates was observed, as seen in Fig. 3b. According to the EDS (in Table 4) and XRD analysis results, the dark area (DA) and the light area (LA) is the same phase (see Fig. 3a–d). It is suggested that the addition of V element can reduce segregation of Al, Ni, Fe, Cr elements and tend to homogenize the microstructure. Table 3 shows the chemical mixing enthalpies of the atomic pairs among the alloying elements. The mixing enthalpies between aluminum and cobalt, nickel, chromium, and iron are -19 , -22 , -10 , -11 kJ/mol, respectively. Therefore, it is concluded that aluminum attracts nickel and cobalt to form the Al-NiCo-rich phase while the rest of elements form the FeCr-rich phase.

3.3. Mechanical properties

The compressive engineering stress–strain curves of AlCoCrFeNi $_x$ alloys are plotted in Fig. 4. The detailed mechanical properties and Vickers hardness are listed in Table 5. It can be seen that the yield strength σ_y increased from 1378.9 to 1726.5 MPa while the plastic strain ε_p decreased from 22.7% to 0.7% with the increase of V content. For V0.2 alloy, the fracture strength σ_b and the plastic strain reached the maximum value of 3297.8 MPa and 0.268, respectively. All alloys were found to have high yield strength, which well indicates the potential of applications. The Vickers hardness curve of AlCoCrFeNi $_x$ alloys is shown in Fig. 5. Interestingly, the Vickers hardness increased almost linearly from HV534 to HV648.8 with the increase of V element. The linear line can be expressed as $Y_{HV} = 114.8x + 534$, where Y_{HV} is the Vickers hardness and x is the value in AlCoCrFeNi $_x$. The increase of yield strength and hardness with the increase of V content is predominantly attributed to the solid solution strengthening corresponding to the shift of the diffraction peaks of (110) to the lower 2θ .

The V0.2 alloy possesses the high strength, hardness and plastic at the same time, which can be explained as follows. There is a fine interconnected spinodal microstructure in the V0.2 alloy, as seen in Fig. 2f. Thus, V0.2 alloy has much higher fracture strength and plastic strain in the AlCoCrFeNi $_x$ alloy system. And the strengthening mechanisms in the alloy system are solid solution strengthening and dispersion strengthening [35].

3.4. Magnetic properties

Fig. 6 shows the magnetization curves of the AlCoCrFeNi $_x$ alloys. Clearly, V0, V0.2, and V0.5 alloys show typical ferromagnetic behavior while V0.8 and V1.0 alloys show typical paramagnetic

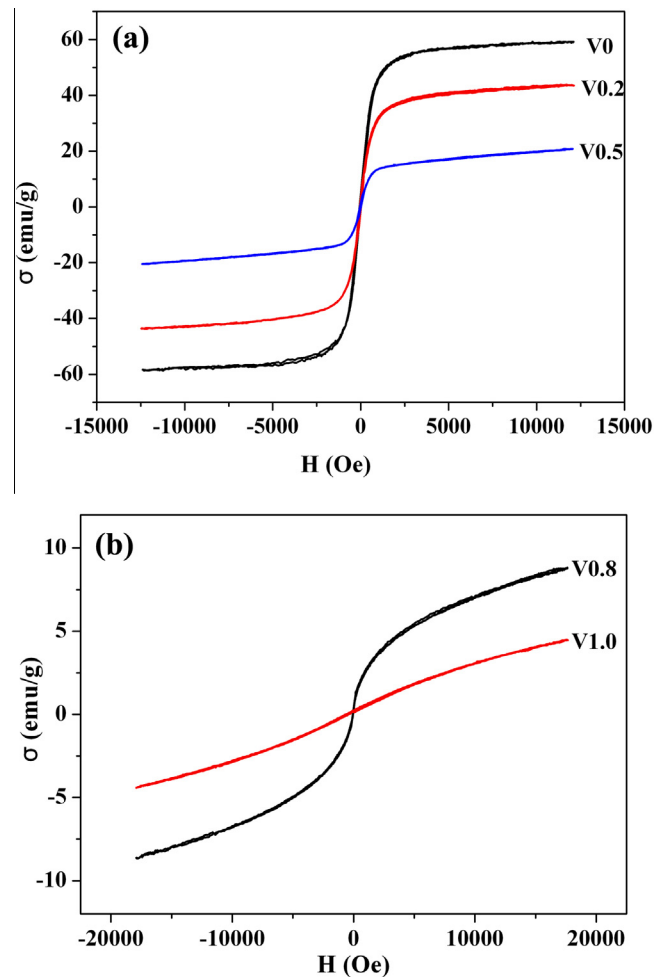


Fig. 6. Magnetization curves of the AlCoCrFeNi $_x$ alloys, (a) $x = 0, 0.2$, and 0.5 , (b) $x = 0.8$ and 1.0 .

behavior, as shown in Fig. 6a and b, respectively. The specific saturation magnetizations of V0, V0.2 and V0.5 alloys under the magnetic field of 12,500 Oe are 59, 43.66, and 20.51 emu/g, respectively. The specific saturation magnetizations of V0.8 and V1.0 cannot be reached under the magnetic field intensity of 18,000 Oe. With the increase of V contents, the transition from ferromagnetic to paramagnetic behavior may be due to the addition of the paramagnetic elements (i.e., Al and V) and the anti-ferromagnetic element (Cr) into the alloy system. And the ferromagnetic elements (i.e., Fe, Co, and Ni) cannot preserve the ferromagnetic property. Moreover, the coercive forces are 36.6, 34.6, 34.8, 31, and 38.1 Oe for V0, V0.2, V0.5, V0.8 and V1.0, respectively, which reveals its soft magnetic behavior. The effect of V on the magnetic property of AlCoCrFeNi alloy is similar to that of Nb element. In AlCoCrFeNiNb $_x$ alloy system [14], when the content of Nb element increases from $x = 0$ to 0.75, all alloys exhibit ferromagnetic property but the permeability decreases from 2.0×10^{-2} to 3.0×10^{-3} . Hence, the further increase of V element may lead to the transition from ferromagnetic to paramagnetic behavior.

4. Conclusions

The AlCoCrFeNi $_x$ (x value in molar ratio, $x = 0, 0.2, 0.5, 0.8$, and 1.0) alloys are produced by vacuum arc melting. All alloys exhibit predominant BCC solid solution structure. The addition of V element can reduce segregation of Al, Ni, Fe, Ni elements and

homogenize microstructure. With the increase of V content from $x=0$ to 1.0, the yield strength increases from 1378.9 to 1726.5 MPa, the hardness increases from HV534 to HV648.8, and the plastic strain decreases from 0.227 to 0.007. A high compressive strength of 3297.8 MPa and plastic strain of 26.8% was obtained in AlCoCrFeNiV_{0.2} alloy. It is found that V0, V0.2, and V0.5 alloys show typical ferromagnetic behavior while V0.8 and V1.0 alloys show typical paramagnetic behavior.

Acknowledgments

This work was supported by The National Natural Science Foundation of China Grant Nos. 51134013 and 51104029, the China Postdoctoral Science Foundation Funded Project, and the Fundamental Research Funds for the Central Universities.

References

- [1] Cantor B, Chang ITH, Knight P, Vincent AJB. Microstructure development in equiatomic multicomponent alloys. *Mater Sci Eng A* 2004;375–377:213–8.
- [2] Yeh JW, Chen SK, Lin SJ, Gan JY, Chin TS, Shun TT, et al. Nanostructured high-entropy alloys with multiple principal elements: novel alloy design concepts and outcomes. *Adv Eng Mater* 2004;6:299–303.
- [3] Yeh JW, Chang SY, Hong YD, Chen SK, Lin SJ. Anomalous decrease in X-ray diffraction intensities of Cu–Ni–Al–Co–Cr–Fe–Si alloy systems with multi-principal elements. *Mater Chem Phys* 2007;103:41–6.
- [4] Fang SC, Chen WP, Fu ZQ. Microstructure and mechanical properties of twinned Al_{0.5}CrFeNiCo_{0.3}Co_{0.2} high entropy alloy processed by mechanical alloying and spark plasma sintering. *Mater Des* 2014;54:973–9.
- [5] Ye XY, Ma MX, Cao YXL, Liu WJ, Ye XH, Gu Y. The property research on high-entropy alloy Al_xFeCoNiCuCr coating by laser cladding. *Phys Procedia* 2011;12:303–12.
- [6] Chou HP, Chang YS, Chen SK, Yeh JW. Microstructure, thermophysical and electrical properties in Al_xCoCrFeNi ($0 \leq x \leq 2$) high-entropy alloys. *Mater Sci Eng B* 2009;163:184–9.
- [7] Zhang Y, Zhou YJ, Lin JP, Chen GL, Liaw PK. Solid-solution phase formation rules for multi-component alloys. *Adv Eng Mater* 2008;10:534–8.
- [8] Guo S, Liu CT. Phase stability in high entropy alloys: Formation of solid-solution phase or amorphous phase. *Prog Nat Sci: Mater Int* 2011;21:433–46.
- [9] Yang X, Zhang Y. Prediction of high-entropy stabilized solid-solution in multi-component alloys. *Mater Chem Phys* 2012;132:233–8.
- [10] Zhang Y, Peng WJ. Microstructure control and properties optimization of high-entropy alloys. *Procedia Eng* 2012;27:1169–78.
- [11] Chen WP, Fu ZQ, Fang SC, Xiao HQ, Zhu DZ. Alloying behavior, microstructure and mechanical properties in a FeNiCrCo_{0.3}Al_{0.7} high entropy alloy. *Mater Des* 2013;51:854–60.
- [12] Zhu JM, Fu HM, Zhang HF, Wang AM, Li H, Hu ZQ. Microstructures and compressive properties of multicomponent AlCoCrFeNiMo_x alloys. *Mater Sci Eng A* 2010;527:6975–9.
- [13] Zhou YJ, Zhang Y, Wang YL, Chen GL. Solid solution alloys of AlCoCrFeNiTi_x with excellent room-temperature mechanical properties. *Appl Phys Lett* 2007;90:181904.
- [14] Ma SG, Zhang Y. Effect of Nb addition on the microstructure and properties of AlCoCrFeNi high-entropy alloy. *Mater Sci Eng A* 2012;532:480–6.
- [15] Chuang MH, Tsai MH, Wang WR, Lin SJ, Yeh JW. Microstructure and wear behavior of Al_xCo_{1.5}CrFeNi_{1.5}Ti_y high-entropy alloys. *Acta Mater* 2011;59:6308–17.
- [16] Hsu CY, Sheu TS, Yeh JW, Chen SK. Effect of iron content on wear behavior of AlCoCrFe_{0.5}Mo_{0.5}Ni high-entropy alloys. *Wear* 2010;268:653–9.
- [17] Lee CP, Chen YY, Hsu CY, Yeh JW, Shih HC. Enhancing pitting corrosion resistance of Al_xCrFe_{1.5}MnNi_{0.5} high-entropy alloys by anodic treatment in sulfuric acid. *Thin Solid Films* 2008;517:1301–5.
- [18] Chen YY, Duval T, Hong UT, Yeh JW, Shih HC, Wang LH, et al. Corrosion properties of a novel bulk Cu_{0.5}NiAlCoCrFeSi glassy alloy in 288 °C high-purity water. *Mater Lett* 2007;61:2692–6.
- [19] Hemphill MA, Yuan T, Wang GY, Yeh JW, Tsai CW, Chuang A, et al. Fatigue behavior of Al_{0.5}CoCrCuFeNi high entropy alloys. *Acta Mater* 2012;60:5726–34.
- [20] Senkov ON, Scott JM, Senkova SV, Meisenkothen F, Miracle DB, Woodward CF. Microstructure and elevated temperature properties of a refractory TaNbHfZrTi alloy. *J Mater Sci* 2012;47:4062–74.
- [21] Tong CJ, Chen MR, Chen SK, Yeh JW, Shun TT, Lin SJ, et al. Mechanical performance of the Al_xCoCrCuFeNi high-entropy alloy system with multiprincipal elements. *Metall Mater Trans A* 2005;36A:1263–71.
- [22] Tong CJ, Chen YL, Chen SK, Yeh JW, Shun TT, Tsau CH, et al. Microstructure characterization of Al_xCoCrCuFeNi high-entropy alloy system with multiprincipal elements. *Metall Mater Trans A* 2005;36A:881–93.
- [23] Zhu JM, Zhang HF, Fu HM, Wang AM, Li H, Hu ZQ. Microstructures and compressive properties of multicomponent AlCoCrCuFeNiMo_x alloys. *J Alloys Compd* 2010;497:52–6.
- [24] Chen MR, Lin SJ, Yeh JW, Chen SK, Huang YS, Tu CP. Microstructure and properties of Al_{0.5}CoCrCuFeNiTi_x ($x = 0–2.0$) high-entropy alloys. *Mater Trans* 2006;47(5):1395–401.
- [25] Lee CP, Chen YY, Hsu CY, Yeh JW, Shih HC. The effect of boron on the corrosion resistance of the high entropy alloys Al_{0.5}CoCrCuFeNiB_x. *J Electrochem Soc* 2007;154(8):C424–30.
- [26] Chen MR, Lin SJ, Yeh JW, Chen SK, Huang YS, Chuang MH. Effect of vanadium addition on the microstructure, hardness, and wear resistance of Al_{0.5}CoCrCuFeNi high-entropy alloy. *Metall Mater Trans A* 2006;37A:1363–9.
- [27] Zhu JM, Fu HM, Zhang HF, Wang AM, Li H, Hu ZQ. Microstructure and compressive properties of multiprincipal component AlCoCrFeNiC_x alloys. *J Alloys Compd* 2011;509:3476–80.
- [28] Zhu JM, Fu HM, Zhang HF, Wang AM, Li H, Hu ZQ. Synthesis and properties of multiprincipal component AlCoCrFeNiSi_x alloys. *Mater Sci Eng A* 2010;527:7210–4.
- [29] Guo JT. *Materials Science and Engineering for Superalloys*. China: Science Press; 2008.
- [30] Zhang Y, Yang X, Liaw PK. Alloy design and properties optimization of high-entropy alloys. *JOM* 2012;64(7):830–8.
- [31] Dong Y, Lu YP, Kong JR, Zhang JJ, Li TJ. Microstructure and mechanical properties of multi-component AlCrFeNiMo_x high-entropy alloys. *J Alloys Compd* 2013;573:96–101.
- [32] Takeuchi A, Inoue A. Quantitative evaluation of critical cooling rate for metallic glasses. *Mater Sci Eng A* 2001;304–306:446–51.
- [33] Wang YP. PhD thesis. Microstructure and properties of AlCrFeCoNiCu multi-principal-element alloys and its composites. School of Materials Science and Engineering, HIT, China. 2009.
- [34] Wang WR, Wang WL, Wang SC, Tsai YC, Lai CH, Yeh JW. Effect of Al addition on the microstructure and mechanical property of Al_xCoCrFeNi high-entropy alloys. *Intermetallics* 2012;26:44–51.
- [35] Porter DA, Easterling KE, Sherif MY. *Phase Transformation in Metals and Alloys*. 3rd ed. Boca Raton: CRC Press; 2009.